

ARTIFICIAL INTELLIGENCE AND ARCHITECTURE

CAMPO-RUIZ

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edited by

Ingrid Campo-Ruiz

KTH Royal Institute of Technology
KTH School of Architecture and the Built Environment
KTH School of Architecture
TRITA-ABE-RPT-2411
ISBN: 978-91-8040-980-3

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Published by KTH Royal Institute of Technology
in Stockholm 2026.
ISBN: 978-91-8040-980-3
TRITA-ABE-RPT-2411
Printed in Stockholm by Universitetsservice US-AB US AB.

Graphic design by Ingrid Campo-Ruiz.



This work was funded by the European Union's Horizon 2020 Research and Innovation Programme under the European Commission Marie Skłodowska-Curie Actions, Individual Fellowship Grant Number 101032687 EQUBUILD, KTH Digitalisation Platform and KTH School of Architecture, Royal Institute of Technology.

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ACKNOWLEDGMENTS

In March 2024, I organized the international, interdisciplinary, and hybrid conference AI and Architecture, at the KTH Royal Institute of Technology in Stockholm, Sweden. Over three days I had the honor of convening 18 experts from different fields across industry and academia to discuss the impact of AI on society and the environment through the built environment. This book creates a framework for discussing AI and architecture with a selection of the thoughts and research presented at this conference by researchers and practitioners.

I would like to express my heartfelt gratitude to everyone who joined and helped with the conference and the book projects. I extend my thanks to all the sponsors who contributed to making this event a reality, in particular, the KTH School of Architecture, and Head of Department Katja Tollmar Grillner, Digital Futures, the KTH Digitalisation Platform, the European Commission, and Marie Skłodowska-Curie Actions.

Special thanks must go to all the speakers: Kutoma Wakunuma, Ricardo Vinuesa, Pablo Miranda Carranza, Emil Björnson, Erik Swahn, David Selander, Andreas Sudmann, Peter Lynch, Mats Nordahl, Heather Suzanne Woods, Sarah Popelka, Wanyu He, Valarie Waswa, Reinhold Martin, Paolo Bottazzini, and interpreter Camilla Sampietro.

Finally, many thanks to the KTH staff, whose support and enthusiasm were instrumental every step of the way. My special thanks to the staff at the School of Architecture and ABE, Digital Futures, KTH Digitalisation Platform, KTH Communications Department, and the KTH Data Protection Officer. I would also like to thank the advisory board. Behind the camera, thanks go to Joakim Nordstrom for managing the audiovisuals of the live event. And thanks to Kim Eddy and US AB for support with the book project.

Ingrid Campo-Ruiz
Stockholm, 2026

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1

PREFACE

Katja Grillner[a]

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In March 2024, I happily welcomed everyone to the first conference at KTH on Architecture and Artificial Intelligence, convened by Ingrid Campo Ruiz. The conference — and this upcoming anthology — provided a valuable opportunity to explore, together with advanced scholars and practitioners, the intersection of these fields and to discuss the transformative potential they hold for architectural education and practice.

For a few years now, artificial intelligence applications have entered our everyday lives and become accessible tools in scholarly and creative work. They offer fast research capabilities, data analysis, writing, visualization, and design support, among other frequently used functions. A few years into this revolution, however, I must confess: I remain a beginner user. I have experimented sporadically — using ChatGPT for text production, assembling basic information, testing out new student assignments, or jokingly trying to provoke its capacity for lies and hallucinations.

The pace of change is dizzying. I find myself

immersed in a sea of shifting realities — picking up new tools, guidelines, ethical and epistemological challenges, and sustainability concerns along the way. I float between inspiring examples of use and more troubling ones.

Of course, I wish I could afford the time to stop, swim ashore, and climb a hill to gain a much-needed (if only temporary) overview — to engage more deeply with current developments in AI technology and architectural design practice. Such engagement is necessary for all of us: to keep up, to stay critical, and to understand the creative and disciplinary implications of these changes and its relevance to the future. It is in this light that I welcome the publication of this anthology.

Now, for the sake of the conference welcome — and again, more than one year later for this essay — I asked ChatGPT: What are the greatest opportunities and greatest risks for architecture and architectural education with AI? This, of course, we can all easily do.

Back in March 2024, I wasn’t necessarily surprised

by the responses. The opportunities listed included design innovation and exploration, advanced analysis and simulation tools, construction optimization, smart buildings, and interdisciplinary collaboration. The last one — interdisciplinary collaboration — was perhaps something I hadn't considered deeply before, and it struck me as crucial. Architecture as a knowledge field will be increasingly challenged as access to spatial design tools becomes widespread and the ability to generate advanced designs without formal training grows. What, then, becomes the architect's true expertise?

ChatGPT also listed risks: over-reliance on AI and digital tools, lack of access to expensive software, data privacy and security concerns, job displacement, and ethical issues related to bias.

Now, fifteen months later — in June 2025 — the response has shifted slightly. Instead of interdisciplinary collaboration, "personalized learning in education" is highlighted as a key opportunity. While interesting, this is not something I have observed yet in architectural education. Among the risks, a new one appears:

curriculum lag — the very challenge I have already confessed to above: The struggle of educational institutions to keep pace with rapid transformations in architectural design and production practices. There is also a stronger emphasis now on the risk of losing foundational design skills, spatial reasoning, and critical analysis — as well as concerns about authorship, accountability, and aesthetic judgment.

In response to those concerns, one might conclude that many of the core activities in highly ranked architecture schools— foundational aesthetic training in spatial and material phenomena, analog drawing techniques, model-making — the messy, material, and sneezy character of the studio space — should probably not disappear… or? Let's see. My current position is that we on the contrary must take great care to defend these pedagogical formats, and continue to engage deeply with analog, tactile, and messy artistic design processes. This is important.

In my view (as an amateur, don't forget), analog training becomes even more relevant in the face of auto-generated complex building designs. If we are

asked to critically assess the qualities of such designs, how do we prepare our students for that task? How do we train aesthetic and critical judgment in this radically new context? How do we assume responsibility, as human agents, for designs generated through machine learning? I may be naïve — even romantic — but we will still live in analog bodies, in material spaces, even as those spaces increasingly extend into virtual realms (and our bodies, perhaps, along with them).

Of course, I am aware that there is a whole universe of generative processes out there — parts of our field that I am myself still ignorant of. We need to embrace and encourage critical and creative explorations of these domains in our programs. We need to train students to move fluidly between slow, analog, tactile artistic design processes and fast, iterative, semi-automated ones — going from making, to assessing, and then back again through disassembling, unmaking, and materially interrogating that which was first generated automatically.

Thank you so much, Ingrid Campo Ruiz, for initiating the conference and this anthology — and for

choosing to focus particularly on ethical issues such as bias and data harvesting. I, too, feel intrigued, excited, confused, and concerned — all at once — by what is unfolding. All of this is happening against the backdrop of a rapidly changing world, with its immense challenges: escalating wars, climate crisis, and the rise of authoritarian regimes. In this light, I believe radical, critical, and creative exploration of current AI technologies is not just important — it is essential.

Katja Grillner

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2

INTRODUCTION

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Artificial Intelligence (AI) is a territory under construction at a planetary scale. AI is zoned, controlled, and unevenly serviced. In this new territory, the building materials are energy, attention, and labor, and they are reassembled as predictions and recommendations.

The interfaces of AI are smooth, but the ground conditions are not: bias, unfair labor, uneven global benefits, among others. Efficiency has become a new form of justification. AI is a territory where spatial decisions are made elsewhere, at high speed, and on a global scale.

This book examines the impact of AI on societies, the environment, and architecture from diverse critical perspectives. The different chapters in this book are explorations in this new territory, discussing how AI shapes everyday choices and political beliefs, how it reproduces gendered norms, and how it rewires our relationship to one another, architecture and the environment.

Architects are often trained to think from a systems perspective, at different scales, and on the social, environmental, and political consequences. AI promises to help us, architects hold more parameters in our hands at once for more foresight, more capacity to understand climate, mobility, biodiversity, and human needs. AI is already in the architectural studio, generating sketches, façades, layouts, and photorealistic futures in seconds. For architecture, AI produces a new genre of knowledge: hyperreal and instantaneous.

Because architecture is interdisciplinary, this book builds bridges into wider AI conversations. If AI is redesigning the conditions of life, architects should learn to work with it critically and with care.

2.1. AI

The full extent of the impact and risks of AI is unknown. The most evident include source bias, videos and images that cannot be differentiated from

reality, untraceable references, and detriment to some members of society .

“But there is worse to come” states The Future of Life Institute, when referring to AI’s risks (Future of Life Institute, n.d.). The institute’s open letter of March 2023, calling for a global six-month moratorium on the development of AI models and alerting about the unknown risks of AI, has been supported by 33,707 signatories (Future of Life Institute, 2023). More recently, the institute has extended that request to a 15-year moratorium on developing artificial superintelligence, which is, in their own words, “likely to be unpredictable, uncontrollable, unalignable and ungovernable by existing institutions”.

Furthermore, other experts have warned about the imminent dangers of AI. AI systems that surpass human intelligence with so-called superintelligence may lead to unmanageable consequences (Rawas, 2024).

The European Commission defines AI systems as machine-based and designed to operate with varying levels of autonomy, which may adapt and

generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments (The European Parliament and the Council of the European Union, 2024). Although AI has been developed over decades, the most recent AI systems represent a turning point with their perceived ease of use (Choung, David and Ross, 2023), hyperrealist images (Miller et al., 2023), friendly conversations (Chaturvedi et al., 2023), and predictions of next words, readings, and drawing lines. AI systems are applied in multiple contexts, from public spaces such as traffic, agriculture, weather forecasting, security, and research, to private aspects of our everyday lives, where they assess our health, looks, private life, finances, consumption, home decoration, and leisure.

Until now, for decision-making, we have largely relied on human-to-human recommendations. However, AI can now mimic certain aspects of human behavior and provide us with recommendations in multiple aspects of our lives. Some people may even prefer interaction with machines rather than with other

humans precisely to avoid social embarrassment and insecurities.

2.2. Architecture

AI has made a quick entry into architecture, arts, and design. AI systems have already been specifically developed to assist architects with generating sketches, designing building façades to optimize thermal load, or speeding up layout plan design (Qian et al., 2023). AI tools can render hyperrealist images of rooms or entire buildings in a matter of seconds. Producing photorealistic images, however, takes human specialists many hours or days. It could be argued that it is a matter of time before AI replaces architects.

I contend that architects can use AI to contribute to this new global scenario; the most exciting question is how? AI systems open up a wide range of new fields of research and work for architects. In design, AI systems can function as assistant tools

that automate repetitive work and as rapid sources of inspiration through image production. In planning, AI systems can help architects consider and understand many more parameters of place and the people who live there than in the past. Moreover, AI systems are also transforming the way people interact with each other. Architects can be key professionals in helping understand these urban transformations in relation to other critical aspects of everyday lives.

However, the places we inhabit are not just rooms, and architecture is not just about having an idea, or making a photorealistic render or an evocative image. Architecture also implies understanding the processes, parameters, possibilities, and most of all, the environment and people. Architects are used to envisioning scenarios within big social, environmental, and economic challenges, helping organizations prepare for complex futures.

This book aims to include architects and the built environment in the discussion on the qualitative impact of AI on societies and the environment, addressing both the risks and benefits of AI. It also aims to tackle

the challenges and potential of AI for architecture. With this book, I aspire to open new areas of discussion that will inspire and empower many.

2.3. Interdisciplinary

Architecture is interdisciplinary by nature. The aim therefore is to build bridges with other fields of knowledge and points of view, to assess AI and contribute to the international debate.

Besides the benefits and risks directly associated with AI, it amplifies many existing societal challenges. The implications of AI may differ in each field. It is therefore essential to integrate multiple disciplines and perspectives to assess AI from a broader perspective, addressing tensions between disciplines and agendas.

An increasing number of tools help evaluate the qualitative aspects of AI systems. The OECD Network of Experts on AI created the OECD Framework for the Classification of AI Systems to help policy-makers and other decision-makers evaluate AI systems' risks

and benefits to inform AI policies (OECD, 2022). This book situates AI in an interdisciplinary context that intersects with architecture.

2.4. Open Questions

This book explores important questions about how we interact with the world around us within contexts that are changing due to AI systems, shifting the focus from the existing discourses by including architecture and its many contexts in the discussion and questioning the notion that AI systems can be generalized.

The questions for discussion are the following:

- How can architects and creators contribute to addressing this emerging concern regarding the development of AI? Will AI replace architects in some tasks, and what paths forward can it open? How is AI affecting research methods in architecture?
- How much are we influenced by AI in our everyday choices, political opinions, travel destinations, and career paths? How is AI shaping

critical discourse? How is AI influencing opinions and perceptions of reality about social, democratic, artistic, and urban domains on different scales?

- How is AI affecting gender and non-binary issues? In what tasks will AI provide substantial help for a more integrative society?
- How is AI shaping our interaction with each other and with the environment?

This book is organized into three main blocks. The first section is titled Unveiling the Potential of AI. In the chapter “AI in the room”, I take the architect’s perspective to address how AI systems can redistribute people in cities and redefine culture, and how AI-generated images relate to the space of the human body, reflecting and helping identify prevailing stereotypes. Some urban areas and AI-generated images can be exploited by the powerful, favoring certain cultural expressions and perpetuating power imbalances.

The second section AI for Sustainable Environments combines architectural and computer science perspectives. In their chapter “Applications of

Machine Learning in the Design of Wildlife Corridors and Urban Wild Gardens", Peter Lynch and Mats Nordahl examine and explain an AI method for monitoring taxa of interest, to guide planning and design decisions.

In her chapter "Bridging Horizons: The Transformative Power of Artificial Intelligence in Shaping an Inclusive Future", Valarie Waswa addresses the regulatory framework of AI for the sustainable development of places, arguing the potential of AI to drive sustainable development by comparing regulatory frameworks in Europe and Africa.

Power Dynamics for Tomorrow, in the third section, interconnects philosophy and the built environment. In his chapter "What stochastic parrots tell us about the world of AI", Paolo Bottazzini argues that large language models such as ChatGPT are not capable of developing planning. In the future, AI systems will be able to implement technically sound projects without understanding the ethical context. Bottazzini warns about the risk of prioritizing technical solutions without

political awareness and public debate.

Regarding the ethical implications of smart environments, in her chapter “AI and the Politics of Power: Implications for Society and the Architecture of Space”, Kutoma Wakunuma argues for a more responsible AI by examining responsible innovation in architecture to address the ethical challenges of AI systems beyond mere efficiency and productivity.

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3

AI IN THE ROOM

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AI IN THE ROOM

Ingrid Campo-Ruiz

Abstract

Cities are built in the collective imagination. Urban spaces get loved, photographed, renovated, and priced out. Artificial Intelligence (AI) has moved into that territory as a daily editor. AI crops, ranks, amplifies. AI decides what counts as a destination.

I call this new condition *synthetic context*: an AI-filtered version of urban space assembled through reels, map pins, generated images, and the dull authority of repetition. Synthetic context arrives before we reach any place: pre-packaged, clean, persuasive, and oddly confident. The city becomes a recommendation field. That field describes aspects of

reality, and it also produces them.

The first effect of AI-driven visibility is redistribution of people, attention, and investment. Some districts become hyper-legible and culturally rewarded. Other areas turn into leftover space, full of life but lacking data. AI-driven visibility has become a spatial force, acting like zoning. Synthetic context is a privatized system of urban guidance operating through corporate infrastructures. Synthetic context rewards the photogenic, the consumable, and the easily monetized. Algorithms just keep recommending, and cities rearrange. AI edits become a new reality.

The second effect is the narrowing of who is imagined to belong in urban space. Prompt “woman” and the output is often young, white, advertisement-ready female faces. These quick visuals slip into mood boards, slide decks, and design references. Exclusion arrives as absence. The city that follows inherits that narrowness.

If synthetic context becomes the default entrance to cities, I propose *frame activism*: intervening in how synthetic context is assembled. Frame activism

means widening what counts as context. Reintroduce plural narratives back into circulation. Combine AI outputs with other forms of knowledge, including lived experience and ethical judgment. Framing is not neutral. Once synthetic context becomes normal, it materializes into habits, access, and, eventually, the futures our cities allow.

Keywords: architecture; AI; spatial justice; equality; urban space

3.1. Introduction

There is a new city power: AI-driven visibility. Cities are built twice, once in construction materials and once in imagination. The imagination shapes what becomes desirable or neglectable. Artificial Intelligence (AI) has taken over that second territory on a massive scale. AI systems set the mood of a neighborhood before you arrive. AI systems turn certain streets into magnets of activity and leave others as blank space. AI increasingly acts as the master editor by cropping, ranking, and amplifying the relevance of urban areas. AI recommends what is worth seeing and what feels safe enough to enter. AI systems decide what becomes visible and desirable.

AI predicts our next move, but it also may help produce it. AI sets the mood of a neighborhood before you arrive. What gets highlighted by AI becomes a destination. What fades from view becomes background.

AI is generating a new urban context built through ranking, repetition, and omission. I call this

synthetic context: an AI-filtered version of urban space that spreads through massive, fragmented recommendations and generated content. The stakes are massive. What gets repeated becomes real, and what falls out of frame becomes easier to neglect, displace, or redesign away. The rules governing these new recommendations and rankings of AI come through corporate systems operating outside democratic accountability. AI sells synthetic context back to us as a commodity.

Some districts become rewarded, while others fade into the background. AI recommendations work as information and also as a direction leading people and investments towards some areas.

AI can be understood as an arrangement of technology, institutions, and ideology that influences knowledge production, social organization, and the distribution of power (McQuillan 2022). There is a solid body of research on smart cities, platforms, surveillance, and gentrification. What is still under-described is how AI's feeds, recommendations, and generated images function as a new context in

practice. This text steps into that gap. I examine how AI systems redistribute power through visibility across the built environment.

Two arguments follow. First, AI-driven redistribution can intensify inequality. Certain districts become hyper-visible, culturally rewarded, and economically extracted. Others become muted and treated as leftover space. That redistribution reshapes culture by amplifying some places and dimming others.

Second, generative images often reproduce stereotypes about who occupies urban space, considering gender and age. I assess AI-generated images in relation to the human body: who appears as the assumed inhabitant of urban space, versus who rarely appears at all.

My claim here is that the practice of architecture needs to evolve if synthetic context becomes the default starting point for what many people see and assume. Urban meaning and imagination are being processed by AI systems, impacting urban diversity and inclusion. Injustices arise when only some areas are rewarded with attention. We need to combine synthetic contexts

with other forms of intelligence, such as local memory, fieldwork, and lived experience. We also need to assess these synthetic contexts to understand what gets pushed out of their frame. I present a way to see AI as a spatial force already at work, shaping urban movement, access, and imagination.

When AI systems shape urban imagination, the consequences materialize in streets, accessibility thresholds, and habits. The new synthetic context becomes materialized, and those new spaces build the future.

3.2. Synthetic contexts

We arrive in urban spaces that already feel decided. Filtered. Liked. Before our shoes even hit the ground, the place has already introduced itself through rankings and reels. The same “hidden gems” reach millions of people at once. Context, once discovered through others, on foot and over time, now arrives pre-packaged by AI (figure 1).

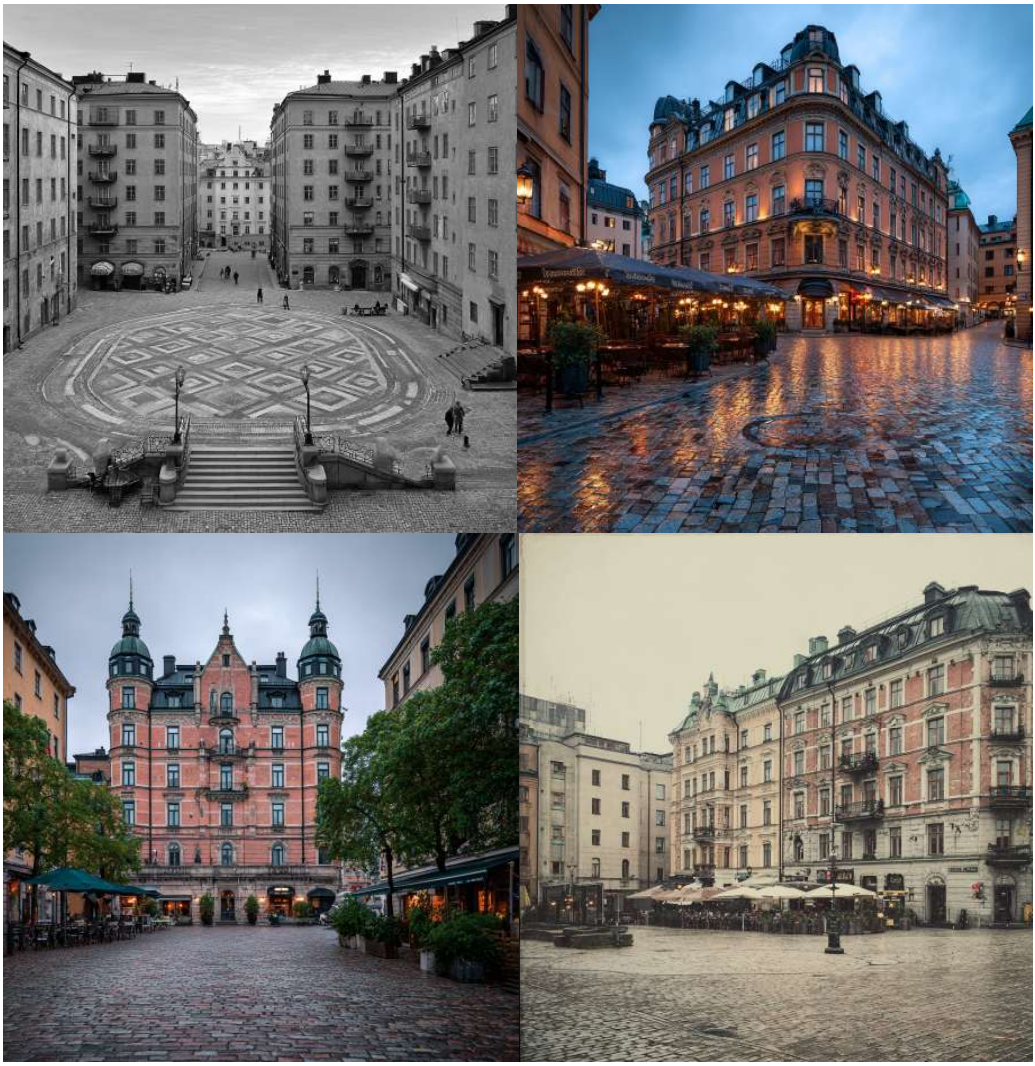


Figure 1. Images obtained interacting with the generative AI bot Midjourney. Prompt: “A square in Stockholm.” (2025).

Context used to be something you earned slowly. You walked it. You got it wrong. You adjusted. Architects turned that discovery into a ritual, with the site visit to understand the “local character.” Context could justify many design decisions. Understanding context felt ethical and stable. An architectural project could sound inevitable, as if the place itself demanded it.

The term context gained popularity right when the ground started shifting. Ernesto Rogers wrote ambiente referring to the lived surroundings, and English criticism translated it into “context” (Forty, 2000). Frampton reclaimed the local context as a key aspect of architecture, including climate, topography, craft, and material culture (Frampton, 2020). Tafuri insisted that architecture sits inside ideology and political economy (Tafuri, 1976). Colomina traced how the media turns buildings into public life through images (Colomina, 1994). Haraway pushed the point further: every view is partial and situated (Haraway, 1988). Context has been imagined in many different ways.

Now that imagination has a new manager. AI increasingly shapes how cities are first seen and entered. AI edits urban life through selection and repetition: cropping, ranking, and amplifying. AI recommendations suggest what is worth seeing and what feels safe enough to enter.

AI's power works smoothly, reaching us as a screenshot from a friend, a hotspot on a map, or a video on social media. Urban spaces look bright and orderly. Then people come. Crowds form. Income follows. A street gains visibility, visibility attracts people, and people attract investment. The chain moves quietly and relentlessly.

Multiple 8-second Instagram posts about the same "quiet" café turn into fifty weekend visitors. Fifty visitors form a queue. That queue convinces the landlord to raise the rent. The new price attracts a new tenant. That tenant imposes their own rules. The district becomes unfamiliar to its previous residents. No masterplan needed. Just posting, again and again, until the city changes.

It all looks ordinary. Over time, this redistribution

shapes culture, access, and investment, turning places into market-facing narratives optimized for attention. The effects vary. Tourist centers become places primed for a new form of value extraction: AI-driven visibility. Peripheral neighborhoods, low-connectivity zones, and urban spaces with thinner digital traces fade in public view. The cultural presence of unhip shops and non-commercial squares diminishes. Their legitimacy starts to feel conditional. Visibility becomes a new kind of zoning, distributing attention, and attention redistributes value.

I argue that AI produces *synthetic contexts* in urban spaces. Synthetic context is the AI-filtered version of urban space assembled through rankings, recommendations, and generated images: fragmented, persuasive, and repeated. Synthetic context needs circulation, habit, the dull authority of repetition. AI recommendations guide movement, investment, and even design inspiration. The city becomes a recommendation field; this field is now a reality. Recommendations appear optional, but in reality, they guide massive numbers of people. Synthetic context

works through persuasion. As synthetic context spreads, the visibility of some ways of living weakens. Their legitimacy starts to feel optional.

If AI systems decide what becomes visible and desirable, they effectively plan cities without transparency. The new urban synthetic context has become a privatized form of urban governance. Synthetic context turns visibility into a new form of urban zoning. Some neighborhoods are consistently amplified while others are consistently dimmed. AI rewards places that perform well as images, pushing cities toward a photogenic, consumable urbanism. Spaces that support slow, unprofitable, everyday life lose cultural priority. The city shifts toward what is easy to photograph, review, and monetize. Context shifts from being a contested, historical reality to a market-facing narrative engineered for consumption. Synthetic context assembles the city through a narrow repertoire of signals. Context has always been woven. Now the loom sits in the cloud, and it runs all day.

3.2. AI and our relationship to body space

Text-to-image generative AI tools feel like magic: you type a sentence, and they generate an image. Now try prompting: “woman” (figure 2). What comes back is mostly a specific woman. Often white and young. Often symmetrical in the way advertisements are.

These images show up as “just an image” but their impact is not trivial. These images circulate everywhere now. They slip into slide decks, websites, mood boards, and leaflets: anywhere a visual is needed fast.

Creativity is often a social phenomenon. We build ideas by borrowing and remixing other people’s ideas. Generative AI accelerates and industrializes that process, influencing who gets imagined.

Architecture begins with the human body. The body is our first room. Everything else follows: the height

of a ceiling, the grip of a door handle, the weight of a window, the width of a threshold, the comfort of a bench, the ease of a staircase.

With generative AI, the “default human” becomes narrow and repetitive, easily guiding the collective imaginary. The actual social diversity is quietly narrowed to a few stereotypes. Exclusion does not only arrive through hostility. Exclusion also arrives through omission.

These AI-generated images are not random. They are the outcome of training systems that rely on thousands of workers who attach labels to images (Denton et al., 2021). Those labels carry values, assumptions, and the moral standards of each worker. A person can be tagged as a “loser”, a “drug addict”, or worse (Crawford & Paglen, 2019). Once written, the AI system learns this tag and repeats it. Therefore, these descriptions inevitably adapt better to the moral standards and cultural perceptions of those who added the descriptions to the photographs, and not necessarily to the rest of humanity.

Even when the language stays polite, the pattern

can be violent. Studies of image generation show older gender stereotypes: fewer depictions of women in male-dominated professions, and women portrayed as agreeable, smiling, and looking down (Sun et al., 2024). Many ML systems underrepresent, misrepresent, or directly exclude women, LGBTQIA+, individuals with special health needs, people with lower or no income, and people with darker skin tones (Denton et al., 2021; Heldreth et al., 2024; Yoder-Himes et al., 2022; Georgievskaya et al., 2023).

So the images from many generative AI systems are not “just images”, they are condensed politics. These images are the crystallization of whoever had the time, power, and cultural dominance to produce data at scale and of whoever wrote the labels that turned that data into training material.

In this regard, the European Institute for Gender Equality defines gender bias as “prejudiced actions or thoughts based on the gender-based perception that women are not equal to men in rights and dignity” (“European Institute for Gender Equality” 2024). O’Connor and Liu believe eliminating all gender biases



Figure 2. Images obtained interacting with the generative AI bot Midjourney. Prompt: “A politically active woman. Photorealistic.” (2025).

in AI is unfeasible due to their strong cultural roots. As an alternative, they suggest regulating evident biases through an interdisciplinary approach (O'Connor and Liu 2023).

Generative AI can be a double-edged sword: it can help give a better understanding of the predominant stereotypes, but it can also perpetuate them. AI systems can help us identify key concentrations of power. The female aesthetic represented in the images I obtained from interacting with the AI model does not mean that women who represent the beauty standards reflected in the AI-generated images hold positions of power. Rather, these standards represent an aesthetic dictated by advertisements and other powerful actors. AI images do not necessarily reflect the majority; they reflect the interest of those who feed the internet with more data entries.

This is where the architectural stakes sharpen. AI now mediates who is imagined as the user of space. When whole groups remain thinly represented in the image-world that architects increasingly draw from, and the city that follows risks inheriting that narrowness. If

AI curates the bodies we imagine, and curates the city we desire, narrowness becomes the norm.

Yet cities never stay still. Urban space changes through relationships, not just through form. Urban space includes networks of people, flows of attention, and informal boundaries. Sometimes a “border” forms without walls, through shared interests, digital signals, and collective behavior. A city square, for instance, can change radically from a marketplace to a demonstration hub, depending on the use made of it.

3.4. Proposed Steps: Frame Activism

Some AI systems are creating artificial contexts that are often partial, commercial, and biased imaginaries of cities with a powerful influence on shaping cultural narratives and early phases of our approach to cities. Images are not mere decorations, but active shapers

of our world.

We have entered a moment when “context” rarely arrives by walking the city. The new synthetic context emerges from scrolling and browsing. AI now assembles places through rankings, images, and recommendations. So the task for architects shifts. The new question shifts from “how do we respond to context?” to “who produces the context, who benefits from it, and who disappears inside it?” When AI erases people, it often does it through absence. These repeated absences become the new reality.

As a response to the deleterious effects of synthetic contexts, I propose *frame activism*: a way of contesting and shaping the machinery through which AI constructs places. Frame activism intervenes in that synthetic context, expanding what enters the frame. Frame activism can mean publishing and reintroducing local histories into the circuit of recommendations, insisting on diversity and plural uses of space. With frame activism, context becomes a site of work again, a narrative that is contested, shaped, and redesigned by many actors. Frame activism treats AI’s “context”

as a political artifact that can be reassembled differently. Frame activism means exposing AI's city-making, working with local groups to bring suppressed knowledge back into circulation, and feeding plural narratives into design processes. Frame activism is a practical response to a new kind of power.

3.5. Conclusions

AI has become a major actor in the built environment. Generative AI tools can intensify inequality by redistributing people through recommendation systems, redefining what counts as culture, and repeating stereotypes until they start to feel normal. Commercial priorities often steer these systems, and the costs tend to fall on people and places with less visibility, less data, and less political leverage.

When AI favors certain cultural forms, it narrows what a place is understood to be. Some forms of everyday urban life become background noise. Some areas are celebrated, extracted, and priced beyond

reach, while others are treated as leftover space.

AI systems are valuable tools for all societies but are currently managed by only a few organizations. AI and architecture can help bring people together; however, they can also displace and segregate people according to the interests of powerful stakeholders. Public spaces and AI are crucial, publicly funded infrastructure, and as such they should be built collectively. For the common good, we should involve different groups of society to influence and benefit from AI decisions.

My takeaway messages are three:

First, providers of AI systems should develop more inclusive AI systems that broaden who and what becomes visible. Providers of AI systems should disclose the limits of their data and the assumptions embedded in their models. A system that guides behavior while hiding its constraints can act like power without accountability.

Second, to widen AI synthetic contexts. Combine AI with other knowledge systems, including social, humanistic, and ethical reasoning. AI tools tend to

measure what is easiest to capture: clicks, reviews, photos, visibility and money. Frame activism begins by widening the inputs and managing more inclusive data, combining AI tools with other forms of knowledge: fieldwork, lived experience, and ethical judgment. Frame activism means to critically use AI as a context-maker, remembering that not all texts are true, not all images are representative, and not all recommendations serve the public interest. Every search, review, purchase, and image upload helps train the system. We all shape the city's visible surface, often without noticing.

Third, policymakers should support research and regulate generative AI to protect everyone: whose narratives do AI systems amplify, whose labor do they monetize, whose spaces do they leave behind?

We are undergoing a revolution, as the uses of AI have grown exponentially. But not everyone has the tools, the time, or the skills to be an active player in shaping how AI influences our lives through the built environment. I call to all those who have the agency, the skills, and the time, to use AI to help others. We

architects can be activists to use AI for the greater good and to have a positive impact on the places we inhabit. AI systems are a huge opportunity to make the world a better place. We should take advantage of AI systems' positive potential for the mitigation of inequalities in built spaces.

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4

APPLICATIONS OF MACHINE LEARNING IN THE DESIGN OF WILDLIFE CORRIDORS AND URBAN WILD GARDENS

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APPLICATIONS OF MACHINE LEARNING IN THE DESIGN OF WILDLIFE CORRIDORS AND URBAN WILD GARDENS

Peter Lynch, Mats Nordahl

Abstract

The authors are developing an AI-based method of biodiversity monitoring that can identify taxa of interest within a study area, over long periods of time. We are working to develop an automated monitoring system based on this method that is inexpensive and robust enough to be implementable by landscape architects, nature organizations, and urban planning agencies. Such a monitoring system would provide long-term bio-inventory data in near-real-time to guide ongoing

planning and design decisions. The method analyses video data continuously gathered by high-resolution video cameras. An AI-based algorithm identifies and tracks moving objects in the pixel data, generating vector lines that trace each moving object. These vector lines are then classified and associated with insect, bird, and ground animal taxa that have characteristic movement patterns. We describe our implementation of this method at the Timescape Garden, an urban wild garden/biodiversity preserve that we and others have created in Norrköping, Sweden. We identify strengths and drawbacks of our current system; describe improvements that are required in order to realize a practical design tool; and give examples of further applications that we are exploring.

Keywords: biodiversity; bioinventory; long-term ecological monitoring; motion analysis; behaviour analysis

4.1. Biodiversity collapse and design responses

Although we are all aware of the reality of ongoing biodiversity collapse, it is worthwhile to acknowledge the scale of the problem. In the period ~1900–2015 more than 40% of 177 studied mammal species lost more than 80% of their range throughout Europe. A similar pattern is unfolding throughout the world (Ceballos, Ehrilch, and Dirzo 2017). Overall insect biodiversity, “understood as a combination of species richness, absolute and relative abundance, and community complexity and diversity,” is undeniably declining (Zattara and Arbetman 2024). In a 2004 study at the Kullaberg Nature Reserve in southwest Sweden, 27% of butterflies and moths resident in the 1950s were no longer found. “Butterflies exhibited a disastrous decline with a loss of 45% of the fauna…

Almost 70% of red-listed species resident in the area in the 1950s were not found in 2004" (Franzén and Johannesson 2007).

Declines in insect and other animal populations are caused by land-use changes, environmental pollution, biological invasions provoked by human disruption, and climate change. All these causes are linked to human activity. Of these, the most common and best-studied causes are urbanization, deforestation, agricultural intensification, and habitat degradation, which includes fragmentation and habitat destruction (IPBES 2019).

Many insects, amphibians, and birds with limited flight cannot traverse man-made obstacles and exposed terrain. Habitat fragmentation creates vulnerable pockets of small populations; species extinction unfolds through many local extinction events (Ihse 1995). Expansion and interconnection of habitat areas—for example, by creating regional wildlife corridors and ecoducts through urbanized areas—is a widely recognized response to habitat fragmentation. There is longstanding agreement that such green

infrastructure projects should incorporate long-term ecological monitoring (LTEM) programs as an integral part of their design and implementation (Beier and Loe 1992). LTEM builds collective knowledge about design for biodiversity, helping to “disentangle change driven by anthropogenic stressors from background noise” (Magurran et al. 2010). It also has the potential to guide specific design decisions related to management and expansion of existing green infrastructures.

As traditionally practiced, architecture, landscape architecture, and urban design lack a feedback loop within the design/construction process. A typical design team bases their design decisions upon experience, precedent, consultation, calculation, modelling, and intuition, but not upon performance data collected at the project site. Design tools that support intuition-guided decision-making by providing feedback about the actual, ongoing consequences of design decisions at the project site could increase the adaptive capacity of the design professions. Some energy-efficient design approaches, for example the Passivhaus design protocol, already require

quantitative feedback. Testing of airtightness during construction is required for Passivhaus certification (BRE n.d.). In an analogous way, landscape design and urban design methodologies would benefit from the incorporation of evidence-based approaches and tools (Berghauser Pont 2017). The authors are developing one such tool: an AI-based method for identifying birds, insects, and ground animals present in natural and urban wild environments in near-real-time, over long study periods. The method builds upon an AI-based approach to motion analysis developed by one of us, Mats Nordahl.

4.2. Methods and approaches to biodiversity monitoring

Design and management of wildlife corridors and nature reserves relies upon accurate, periodically updated surveys of animal taxa. Planners and designers need to identify which species of concern are present in their study areas. Decisions about

the creation, expansion, and interconnection of habitat areas should be strongly guided by such data (Berghauser Pont 2017). Ideally, surveys would be performed over large geographical areas, over long time periods. Data collection and analysis should ideally function continuously, as with weather stations. Only an AI-based approach is capable of processing such large data streams.

Different research projects are underway to develop such a monitoring and analysis infrastructure (Van Klink et al. 2022). One example is the German AMMOD project, an initiative to develop automated biodiversity monitoring stations, analogous to weather stations (Wägele et al. 2022). AMMOD takes a multimodal approach. The initiative proposes a network of complex sensor stations, deployed at regional, national, or continental scale, that are capable of automatic image recognition of birds, insects, and mammals; acoustic identification of various species; chemical analysis of smells in the landscape; and DNA identification of flying insects, pollen, etc. Other researchers have instead emphasized the need for

denser biomonitoring networks consisting of simple passive sensors (Reboud et al. 2021).

The visual detection of animals in such systems is typically based on tracking algorithms that use state-of-the-art object detection algorithms, e.g., different versions of YOLO (Jiang et al. 2022) or R-CNN (Girshick et al. 2014) to detect and identify objects in each video frame. The object detections are then connected into trajectories by other algorithms, for example using prediction of motion based on algorithms such as Kalman filtering.

These algorithms are often developed with other use cases in mind, such as collision avoidance in autonomous vehicles, and may be less suited to tracking small, fast-moving objects such as birds or insects where less visual information is available in each frame. They may not work as well in situations with noisy and incomplete information, for example using night vision for monitoring in ecological applications, which is necessary because many animals are primarily active during the dark hours.

Our algorithmic approach differs from these

standard approaches since it to a larger degree considers properties of the object's motion rather than appearance—"behaviour analysis" as opposed to "image analysis"— to perform identification. In this way it resembles the visual system of some animals, as well as human peripheral vision, which is particularly sensitive to moving objects.

In the system we have built, continuous video recordings in HD or 4K resolution from multiple stationary cameras are transferred over a network, stored on remote hard drives, and analysed offline. An important aspect of our project, which is currently being implemented, is elimination of the need to store or transmit any image data. Every video feed will be processed at the camera source ("edge computing") and the output will consist only of trajectories of animals. This approach requires simplified versions of the AI algorithms in order to run on less complex hardware developed for embedded systems.

Movements recorded in the video data are converted into vector trajectories. Objects within the camera's field of view are tracked to a high degree of

precision. Using machine learning, the program sorts these vector lines by their properties, which in turn can be correlated with the characteristic movement patterns of different animal taxa. The high-resolution system initially detects all kinds of moving objects, including leaves and branches moving in the wind, falling snowflakes, clouds, shadows, and large moving objects in the background such as vehicles. These are distinguished from animate objects through filtering algorithms based on machine learning algorithms that consider both motion characteristics and static visual characteristics. Detecting small birds in a background of movement in trees is a challenging research problem which is only partially solved.

After this filtering, resulting vector lines are printed or displayed as an overlay drawing of multiple movement traces (figure 1). This visualization of animal activity is reviewed by a human analyst, who, drawing upon prior experience, identifies traces generated by taxa of interest. The human analyst assigns individual vector lines an attribute—the species that most likely generated them—and thereby builds the data set



Figure 1. Vector lines tracking nocturnal movements of hares in Timescape Garden's forest glade in March 2023. Vector lines were isolated by filtering for apparent speed, and then by anomaly detection. Background movement of trees shown in grey. (Image credit: Mats Nordahl, 2023)

required to train a fully automated AI-driven analysis, the project's long-term goal.

The result of such an automated analysis could be formatted as tabulated data or as a tagged version of the vector-line drawing described above. The latter would be particularly suitable for review by designers, who are trained to read drawings as traces of reality, more valuable than tabular information. If the designer responsible for ongoing work at the project site reviewed such drawings periodically as part of the design process, they would gain feedback about changing animal populations—and learn to correlate their ongoing design decisions with the direct and indirect effects of their actions. Objective feedback would help train design intuition.

The design process could also benefit in another way. By periodically reviewing animals' movements in near-real-time at the project site, the designer would constantly be remined of the purpose and consequences of their decisions, and the reality of the situation at hand. Their knowledge and practice could become more situated (Stengers 2018).

Our motion-analysis-based approach promises certain practical advantages over existing approaches to biodiversity monitoring. It should make it possible to identify taxa within a larger study area, using lower-resolution data. It requires less expensive and sophisticated equipment than Scheimpflug lidar approaches to insect detection and analysis (Brydegaard et al. 2021) and can operate over longer periods. Like Scheimpflug lidar, our method can detect and track very large congregations of objects (flocks, swarms, human crowds) and fast and small objects (birds and insects). It can also track camouflaged and hard-to-detect animals. It accommodates the poor visual conditions that are often encountered in field applications (night vision, snow, and rain). Its ability to trace small and fast-moving objects has recently been compared to a range of state-of-the-art algorithms for tracking small objects, with encouraging results.

4.3. Field testing and development at the Timescape Garden, Norrköping

Depending upon how the camera is mounted, the method we are developing can detect and identify birds and larger ground animals within a large study area; butterflies and moths as well as birds and larger animals in a more constricted study area; and even the smallest flying insects within a narrowly focused field of view. We are testing all three configurations at the “Timescape Garden,” a wild urban garden/biodiversity preserve that we, along with other collaborators, have created in Norrköping, Sweden (Lynch 2023).

This study area, a 6000 square meter public garden, is located at the western end of Sylten ridge, a municipally-owned green area directly east of Norrköping’s historic city centre. Three 4K video cameras have been recording the site since spring 2021. From October 2021 to September 2024 the



Figure 2. Plan of the Timescape Garden. Both human and animal paths are represented. Thick lines represent movements of hares and pheasants in April 2021, Nov 2021, and Nov 2022. (Image credit: Peter Lynch, 2023)

design and realization of the garden, as well as our data collection and analysis work, was by Vinnova, Sweden's innovation agency. Norrköping Municipality provided substantial in-kind support. One of us, Peter Lynch, was the project leader.

Before the Timescape Garden was located at the project site, this part of Sylten ridge was semi-maintained and rarely visited. Today, after extensive site improvements, the place attracts more than five times as many human visitors and many more ground animals, insects, and birds (figure 2). The functionality of the garden as a biodiversity preserve—its ability to attract diverse insect and animal populations—was immediately apparent. In June 2022, less than one month after we introduced new substrate and plant beds in the wildflower garden area, a red-listed insect species, the red-tipped clearwing *Synanthedon formicaeformis*, was sighted there by naturalist Robert Petersen, consultant for SWECO Norrköping. According to SLU's open-source database Artportalen.se, this species has only been sighted three times in Norrköping municipality.

The garden is composed of two adjoining parts: a wildflower garden with 215 native and non-native flowering species introduced by plant specialist and garden designer Peter Korn, and a forest glade with over 40 introduced fruit, seed, and nectar-bearing plant species designed by ecosystem restoration specialist Linnea Andersdotter Rundgren. Besides designing the garden and expanding animal habitat, the project's interdisciplinary design team—Peter Lynch, Norrköping municipality's strategic planner Martin Heidesjö, and Swedish dancer/choreographer Anna Asplind, in collaboration with Korn and Rundgren—created new paths, benches, terraces, and seating areas for human visitors.

A video camera located on the roof of a nearby disused power plant, 200 meters from the Timescape Garden, continuously records an overview of the site (Figure 3). We are processing this video data to track birds and analyse human visitors' use patterns. A medium-range camera mounted on a tree in the forest glade captures the movement of small birds and ground animals. A downward-pointing camera with a

narrowly focused field of view was re-mounted in the wildflower garden in late summer 2024. To-date we have collected more than 40 TB of video data.

4.4. Limitations of our current methods of data collection and analysis, and solutions under development

Continuous collection and management of this amount of data, gathered over several years, poses significant technical and logistical problems. It was not practical to transmit video data via the internet. Wi-Fi does not have sufficient range to cover our field locations, and the amount of data transmission excludes wireless networks. Consequently, each camera is connected by an ethernet cable to a dedicated computer system that stores video files on an external hard drive. This means that every data collection station has three points of vulnerability: the camera, the cable, and the computer system. Over the

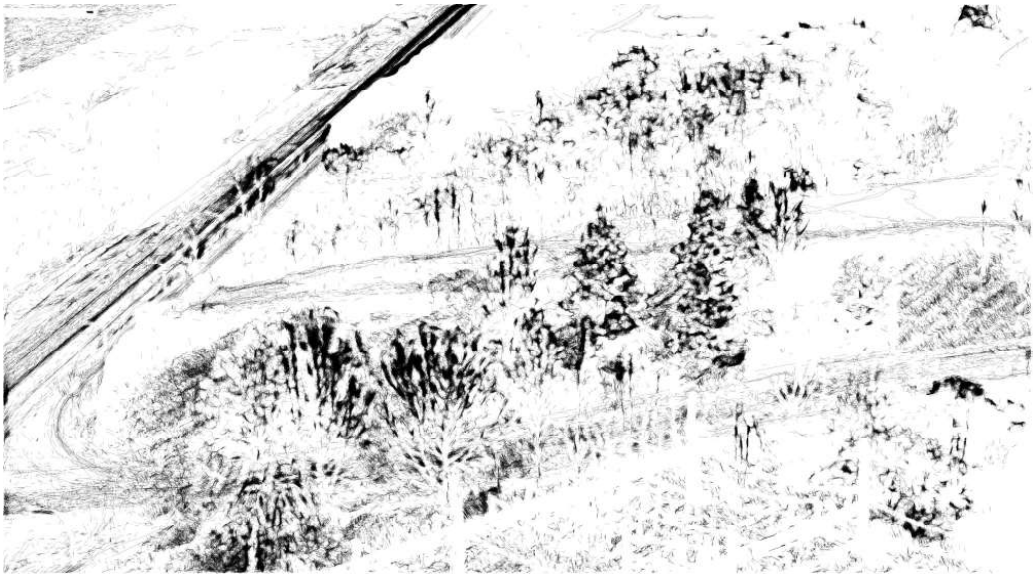


Figure 3. Vector lines tracking movements of people, vehicles, and trees at the Timescape Garden in Feb 2022. Image was generated from video data collected by a camera 200 meters from the project site, filtered for apparent speed (Image credit: Mats Nordahl, 2023).

three years of the project, four cameras have failed or been vandalized, two cables have been cut, and two computer systems have been stolen. Access to data systems has occasionally been blocked due to the presence of asbestos, an arson attempt, and other calamities. Power supplies are repeatedly interrupted; we must check the computer systems every two or three weeks so they can be repaired or rebooted if necessary. Hard drives have proven to be fragile and expensive, but cloud storage is difficult because of bandwidth issues.

Analysis of 40TB of video data poses even more complex practical problems and demands on processing power. Any motion tracking algorithm needs to process raw video data. This means that files must be decompressed before analysis, which requires significant processing power for high-resolution video. The method we are developing will only become practical once the approach to data collection is more robust and data processing is more streamlined. As noted above, we are developing an edge-computing solution that addresses these issues. A GPU based

module mounted beside the camera will convert the video stream to classified motion trajectories. The resulting files are small enough to be transmitted in near-real-time via data upload, and the resulting database is small enough to be stored in the cloud. By eliminating any storage or transmission of image data the system can also potentially avoid privacy issues related to surveillance, a consideration of particular importance in parks and urban wild environments often frequented by humans.

4.5. Additional applications of our method

Identifying non-native plant species that are food sources for native insect species

There is disagreement about the advisability of introducing non-native plant species to gardens and wild nature areas, as we are doing in the Timescape Garden. Plant specialist Peter Korn, responsible for

the wildflower plantings, is often asked about the potential for invasive species, and argues that his planting approach precludes such a risk (Korn 2012). Another common criticism of introduced non-native plant species is the claim that they displace plants that serve as a food source for specialized native insects (monophages). This objection can be challenged, anecdotally, by on-site observations of increased insect populations in the Timescape Garden, including the red-listed insect species noted above. It can be addressed in a rigorous way by gathering more information about insect behaviour. As noted above, we are mounting one of our video cameras on a utility pole within Korn's wildflower beds. The downward-pointing camera will document the frequency of insect visits to non-native plant species in the field of view. In this way we can begin to accumulate data about the attractiveness of specific non-native plant species for specific native insect species.

Mapping migratory pathways and flight paths of bird species

One of our video cameras is located on the roof of a disused power plant, 200 meters from the Sylten ridge and 60 meters above the ground. Many raptors and waterbirds pass through the intervening airspace. A second camera mounted on the power plant roof will allow us to collect stereoscopic image data and trace bird flight paths in three dimensions. This will improve the capacity of the method to identify bird taxa, and could also allow researchers to chart migratory pathways and flight paths of species of concern. Such a study could extend over many seasons and be performed in an economical way. Potential applications include analysis of bird flight patterns in areas where wind turbines are proposed, and estimating the functional connectivity of neighbouring bird habitat areas.

Intelligent landscape lighting

We are also seeking to develop and employ our motion-detection and analysis method as an

intelligent control system for outdoor electric lighting in forested urban areas. The goal is to minimize the negative impact of anthropogenic light at night on other species. Urban green areas like Sylten are constantly visited by both people and animals—who have conflicting requirements regarding illumination. Humans gain safety and security when paths through forested areas are illuminated, but animals benefit from darkness. If a lighting control system were able to discriminate between the nighttime presence of humans and animals, and track and follow their movements in a forested area, it could adjust lighting levels and locations accordingly.

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5

BRIDGING HORIZONS: THE TRANSFORMATIVE POWER OF ARTIFICIAL INTELLIGENCE IN SHAPING AN INCLUSIVE FUTURE

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BRIDGING HORIZONS: THE TRANSFORMATIVE POWER OF ARTIFICIAL INTELLIGENCE IN SHAPING AN INCLUSIVE FUTURE

Valarie Waswa

Abstract

Artificial Intelligence (AI) is a transformative technology that can exceed human limitations and significantly impact society. This chapter explores AI's potential to drive sustainable development, through a comparative analysis of its progress and regulatory landscapes in Europe and Africa.

AI can enhance governance, healthcare, and economic opportunities. In governance, AI can

improve efficiency through data analysis, automate routine tasks, and boost citizen engagement. AI could expedite administrative functions, swiftly analyze medical data, and offer personalized patient support in healthcare. Economically, AI has the potential to generate millions of jobs, such as data scientists, AI ethics officers, and robotics engineers, while boosting and promoting entrepreneurship through innovative products and services.

Europe is setting the pace when it comes to AI development and regulation with forward-thinking initiatives such as the Artificial Intelligence Act and the EU Innovation Package. These not only push for ethical AI practices but also provide much-needed support to small businesses trying to navigate the tech landscape. Meanwhile, Africa faces its own hurdles such as weak infrastructure, limited digital literacy, and disjointed regulations. While a few countries like Kenya, Nigeria, and South Africa are making strides, more still needs to be done. For Africa to truly tap into AI's potential, we must see more unity and collaboration across the continent.

To address these challenges, the chapter proposes several recommendations. These include fostering global collaboration, investing in AI education, creating a unified AI regulatory framework in Africa, and upgrading digital infrastructure. These aim to ensure that AI technology is used ethically and effectively and helps bridge the gap between developed and developing regions.

Keywords: Artificial Intelligence, sustainable development, governance, healthcare, economic opportunities, regulation, global collaboration, digital infrastructure.

5.1. Introduction

We, as human beings are very interesting creatures- we are limited in our strength. We get tired, and we have emotions, that at times cloud our judgment and rationality, and this limitation at times advances unconscious bias, which often leads to continued equality. Yet, we are the smartest living creatures to exist in the universe; so smart that we invented Artificial Intelligence, something bigger than us, an invention that goes beyond our human limits. An invention that has the potential to shape our futures and leave a lasting impact on society.

In this Chapter, we shall delver discussing and analyzing the potential of AI in Promoting sustainable development, while factoring in the current reality, with a comparative analysis of Europe and Africa, and conclude with giving proposed solutions to ensure we

maximally leverage Artificial Intelligence technology to promote a thriving society for current and future generations.

5.2. The Potential of AI in Promoting Sustainable Development

First and foremost, AI can promote better governance of nations. It can do so by boost efficiency, enhance decision-making and promote transparency. It can do so by analyzing data to ensure that policies being made are evidence-based, assisting governments better address societal issues. Further, AI can lead to better administration by automating mundane tasks such as data entry and processing e.g. of citizen information, financial records and administrative documents. AI-driven chatbots and virtual assistants can handle routine inquiries and provide assistance to citizens on various government services, such as applying for permits and licenses. Also, AI can enhance citizen engagement through predictive analytics, which allows governments to

anticipate citizen needs and preferences based on historical data, enabling proactive service delivery.

Secondly, AI has the prospect of promoting efficient healthcare services. Time is really of the essence matters healthcare service delivery. It is quite literally a matter of life and death, and a lot is at stake. AI has the ability to speed up things, by automating administrative tasks such as appointment scheduling and medical record documentation. More than that, AI-powered tools can quickly analyze medical images and patient data rapidly, significantly reducing treatment delays. Also, virtual health assistants can offer personalized support to patients, advancing accessibility.

Another potential of AI is in creating countless income generating opportunities. When considering the broader implications of AI, it often circles back to the fundamental aspect of financial empowerment. In essence, AI can unlock countless avenues for income generation, thereby fostering economic growth and prosperity for individuals and communities alike. The fear and concern that some people have that emerging technologies like AI will make many

jobs extinct and irrelevant is not completely true and proven. This is because while indeed AI may automate most routine and mundane tasks, it has the capability to create new careers and jobs- jobs of the future; jobs with better security and better pay. These jobs include: data scientists; AI ethics officers; AI trainers who are responsible for training and fine-tuning AI models by providing annotated data and feedback, ensuring optimal performance and accuracy in various applications, from natural language processing to computer vision; virtual reality experience designers; human-machine teaming specialists; AI product managers; robotics engineers; and computational linguists. These are just a drop in an ocean of over 97 million jobs that AI has the potential to create by 2025, according to the 'Future of Jobs' Report by World Economic Forum in 2020. When AI is maximally leveraged, unemployment will be a thing of the past.

And it does not end at that- AI does not only have the prospect of creating jobs, but also of boosting entrepreneurship for those with business acumen. How so? Entrepreneurs can create more innovative

products and services by integrating AI capabilities, such as personalized recommendations, predictive analytics, and natural language processing into various industries, from e-commerce to healthcare. Further, entrepreneurs can leverage and rely on data analytics to gain valuable insights into market trends, customer preferences, and operational efficiencies, hence creating a level-playing field for small businesses. Moreover, AI algorithms can analyze financial data to assess creditworthiness and risk, expanding access to credit and financial products for underserved populations.

AI also has the potential to boost the Architectural sector and help promote an inclusive future. It can assist architects to design sustainable structures that minimize environmental impact. Further, AI-driven simulations and modelling techniques enable architects to optimize building designs for accessibility, accommodating the diverse needs of persons and ensuring true inclusivity in architectural spaces.

5.3. Current Reality on Advancement and Regulation of AI

Now that we have seen the expanse of what AI can do for our societies, let us discuss the current reality of how much progress (or not) has been made in leveraging AI technology.

5.4. A Keen Focus on Europe

Given Europe's leading role in AI development and regulation for ethical and responsible AI use, we shall explore the current landscape of AI use, innovation, development and regulation n Europe, from a European Union's standpoint.

On 13th March, 2024, the European parliament approved the Artificial Intelligence Act, being the world's first statutory legislation to regulate AI. Europe is now a global standard-setter in AI. There is a lot to learn from this advancement and borrow a leaf or two. This groundbreaking Act seeks to ensure that humans are at the center of the deployment of AI. This means

making sure that AI benefits society and respects human rights. It aims to prevent any harm that AI could cause and ensure that AI is used in a fair and accountable way. Example of fundamental rights the Act seeks to preserve in the use and deployment of AI include: the right to privacy, matters personal data protection; and the right to equality and freedom from discrimination. The importance of having the right to equality and freedom protected and respected in the use of AI is because AI algorithms may perpetuate bias and unfair treatment based on factors such as race, gender, or socio-economic status, just to mention a few.

Some of the key provisions of the Artificial Intelligence Act are that it has placed a ban on harmful and acceptable AI practices, categorizing AI technology as high, medium, and low hazard, and has different sets of standards and requirements to be met by each category, with ‘high-categorized’ AI technology having more compliance requirements, as they have higher ethical risks.

The AI Act offers a unified and comprehensive

regulatory framework specifically tailored to the European context, ensuring consistency and coherence in AI governance. This regional approach allows for better coordination and cooperation among member states, fostering a harmonized approach to AI while promoting innovation, competitiveness, and trust in AI technologies within the European market.

Another key feature of the AI Act is its binding nature. Statutory acts are usually binding in nature, this being one of those, with sanctions, unlike policies. Failing to comply with certain AI practices can lead to fines up to 35 million EUR or 7% of a company's annual revenue. Other infringements can result in fines up to 15 million EUR or 3% of annual revenue. Providing false or misleading information can incur fines up to 7.5 million EUR or 1% of annual revenue. The severity of fines will be based on factors such as the nature of the violation, the size of the company, and any past violations. Member states are required to report annually to the Commission on the fines they have imposed.

Another key development of European Union in

AI technology deployment and use is the European Union Innovation Package. On 24th January, 2024, the European Commission launched the 'Innovation Package', an initiative that seeks to help small businesses and startups in Europe create AI technology that is reliable and follows EU principles and regulations. Some of the measures in the package include: The 'Large AI Grand Challenge'- a prize that seeks to give AI startups financial support and supercomputing access; the 'AI Factories Set-up'- that seeks to bring together the 'raw materials' for AI, i.e. computing power, data, algorithms, and talent, enabling them to develop the most advanced AI models and industrial applications; and lastly the financial support of public and private investments of around 4 Billion EUR until 2027, specifically dedicated to Generative AI.

Another key development of AI in Europe is the 'AI Skills Strategy for Europe'. This was developed by the Artificial Intelligence Skills Alliance (ARISA) project consortium. The Strategy aims to address skills shortages and mismatches in the AI landscape,

fostering diverse skillsets among stakeholders for systematic impact. It also establishes a framework for enhancing AI education and training across Europe, prioritizing innovation, inclusivity and ethical principles.

Lastly, European Union also established the 'European High Performance Computing Joint'. This is the data infrastructure aspect of AI development in Europe. It is a legal and funding entity, created in 2018, and located in Luxembourg, to lead the way in European supercomputing. Supercomputing resources are extremely powerful computers that can process and analyze large amounts of data very quickly. They are used for complex tasks like simulations, scientific research, and artificial intelligence development. This entity supports the development of key HPC skills for European science and industry.

5.5. A Keen Focus on Africa

It is truly inspiring to witness the great strides and milestones that Europe has taken in advancing the development and regulation of AI. Unfortunately, this

is not the case in Africa. We shall now deep dive into the current reality of the status of AI development and regulation in Africa

Africa is a continent that is made of up of 54 countries. What divides us are the borders and the slight geographic locations. Other than that, we (Africa) have similar challenges, culture, language, legal and political systems. Yet we do not have a unified legal framework to govern AI across Africa, or even a specific regional bloc, such as the East African Community (EAC). I come from the school of thought that Africa should integrate more and more, socially, economically, and hopefully, eventually, even politically, to great degree. Here are some of the African countries that have some regulatory frameworks on AI and development-initiatives on the technology:

A. Kenya

On 8th April, 2024, the Ministry of ICT and Digital Economy launched the 'FAIR Forward- Artificial Intelligence for All' project. The initiative seeks to emphasize the need to develop AI responsibly with a

view of promoting sustainable development in Kenya. What is unique about this project is its multistakeholder approach. It seeks to bring together academia, industry, government, and civil society stakeholders to achieve the collective goal. There will also be working groups handling thematic and focused areas such as Data and Legal Governance. The project seeks to uphold fundamental rights of Kenyans such as right to privacy (data privacy). Beyond regulation, the project aims to foster innovation of AI technology, by investing in AI building blocks, such as computing resources, talent and skills development, and data infrastructure. Kenya being a very climate conscious country, a key consideration of the project is also ensuring that AI development addresses climate change concerns.

Further, Kenya came up with the '2022-2032 Digital Masterplan', a policy that emphasizes the importance of AI technologies, recognizing their growing significance over the next 5-10 years. It seeks to develop an AI master plan to foster research, development, and deployment of AI solutions for local challenges and global export. The plan also highlights

the need for international partnerships to facilitate technology transfers and attract foreign investments in emerging technologies.

B. Nigeria

The National Center for Artificial Intelligence and Robotics (NCAIR) was commissioned in Abuja on the 13th November 2020 as a response to the directive for all agencies under the Ministry of Communications and Digital Economy to formulate practical strategies for enhanced implementation of the digital economy, nurturing AI research and development within the nation.

C. South Africa

On 30th November 2022, the Department of Communications and Digital Technologies (DCDT), the University of Johannesburg (UJ) and the Tswane University of Technology (TUT) established the ‘Artificial Intelligence Institute of South Africa’. This institute aims to focus on research and development, as well as implementation capabilities in AI. In this

way, the institute will help enhance investment in human capital and in the applications that improve collaboration between humans and machines.

Sadly, the current reality is that there are still underlying challenges cross-cutting most African countries that limit the uptake, development and deployment of AI technology. These include:

The first setback is low infrastructure levels and data ecosystem levels. AI services require strong connectivity for real-time data processing and analysis, which is hindered by unreliable or slow internet connections. As a result, the potential benefits of AI technologies are not fully realized in Africa due to infrastructure limitations. Further, insufficient data management rules make it hard to gather and handle data properly. Without clear guidelines, it is difficult to collect the right data needed for AI technology to work well. This means that AI applications, for instance those focused on improving healthcare or farming, cannot be as effective in Africa because they do not have enough data to learn from.

Apart from that, the expensive nature of mobile

internet data and home broadband connections restricts the accessibility and uptake of AI services. High poverty levels and limited purchasing power among the African population exacerbates the issue. Additionally, monopolies or limited competition in the telecommunications sector can lead to inflated prices for internet services.

Another huge setback is huge skills gap levels among most Africans. The current digital literacy levels in Africa are unfortunately very low. The Digital Skills Gap Index 2022 scores African countries between 1.8 and 5. This is below the global average of 6. This is a cause for concern. Low digital literacy levels impede AI adoption by limiting people's understanding and use of AI technologies. There is a general lack of awareness of the benefits of technology in most rural communities in Africa, which makes most people, especially women, reluctant to learn digital literacy skills, let alone more sophisticated skills like the use of AI technology. The skills gap is greater among women, and this is aggravated by underlying general low literacy among women; poverty; and some harmful

cultural norms that propagate gender inequality. All these blocks out women from taking up high-skilled, high-paying AI jobs, perpetuating income disparities and career inequalities.

The final challenge is the issue of labor exploitation within the AI industry. In Kenya, much of the work surrounding AI systems is gig-related, characterized by temporary, project-based engagements. This type of employment poses significant challenges for digital gig workers, as they often lack sufficient rights and protections. The temporary nature of their work means they are not afforded the same legal rights as permanent employees, leading to vulnerabilities and exploitation.

A further complication arises from the remote nature of this work, as many employers operate from different jurisdictions with varying and sometimes conflicting employment laws. This regulatory clash can exacerbate the precariousness of gig workers' conditions, leaving them uncertain about their rights and protections under differing legal frameworks. Consequently, this regulatory inconsistency makes

it difficult to ensure fair treatment and adequate protection for gig workers in the AI sector.

5.6. Proposed Improvements

There are a number of recommendations that when implemented by different actor players, can help us better leverage the potential of AI technology in our societies.

First of all, global collaboration should be greatly embraced. In today's interconnected world, what happens with AI in one place affects us all. For example, if Africa makes strides in AI, it stands to benefits countries across the world, and vice versa. Developed and developing nations should foster partnerships that promote knowledge exchange that boosts AI development. Sharing successful strategies in education, regulation, infrastructure development and labor protection and development empowers nations. Through collaboration, developing countries can leverage the ability of AI to foster growth and address global challenges such as climate change

and unemployment collectively.

Secondly, there should be more investments in AI education and training. This requires dedicated funding from government budgets to develop AI curricula and train instructors. Additionally, supporting grassroots initiatives through grants or subsidies can complement government efforts by providing accessible AI education opportunities at the education level. By combining these approaches, we can ensure a comprehensive and inclusive strategy to bridge the AI skills gap and empower individuals with the necessary knowledge and skills for the future.

Another proposal is to have a unified AI regulatory framework for Africa. This will ensure streamlined governance, boost innovation and cater to the cross-border nature of AI technologies. Further, having a unified regulatory framework promotes trust, facilitates collaboration, and ensures ethical AI practices. It also prevents fragmentation which creates a conducive environment for AI development and investment.

The last (and definitely not least) recommendation would be to upgrade digital architecture in Africa.

Infrastructure is everything when it comes to the development and uptake of AI. It is the backbone of AI. It is therefore important that we are very intentional about investing in better fiber optic cables, satellite technologies to improve internet connectivity especially in Africa, as well as investing in initiatives that improve access to digital devices. It is also important to invest in good and reliable electric supply, which powers AI technology, as well as dataset factories, which is the engine of AI technology.

5.7. Conclusion

In the famous words of Former United States of America President, Mr. John F. Kennedy, he said: “Ask not what your country can do for you- ask what you can do for your country. What do you, as an individual seek to do from this day forward to ensure we leverage the power of AI in an ethical and responsible way, to better achieve sustainable development in our societies. Do not despise your small intervention in achieving the greater goal of digital development through AI. We

need each and everyone of us to play their role, and only then, can we truly realize the fullest potential of AI technology.

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6

WHAT STOCHASTIC PARROTS TELL US ABOUT THE WORLD OF AI

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WHAT STOCHASTIC PARROTS TELL US ABOUT THE WORLD OF AI

Paolo Bottazzini

Abstract

The success of ChatGPT-3.5 has caused the institutional media to spread prophecies about the imminent arrival of the first generation of AGI, Artificial General Intelligence. The expectation that superintelligences of this sort will be able to solve all of the world's problems is associated with great hopes and catastrophic fears. With the aim of empirically measuring the proximity of generative AI to AGI, in March 2023 I formulated three prompts for

ChatGPT-4 to write an essay on artificial intelligence. The errors made by the software offer a framework capable of quelling excessive optimism and alarms. The stochastic parrots generated by ChatGPT-4 demonstrate that current AIs do not have a world (in Heidegger's terms): the AI does not even suspect that the symbols it is manipulating refer to some reality, of which it consequently has no knowledge. Therefore, it is not capable of developing any planning.

The results suggest some reflections. If we consider the development in the artistic field, artificial intelligence seems to be able to identify a sort of "latency" in the works and styles of the past and bring it to light in violation of the Kantian assumption of the Critique of Judgment, according to which the aesthetic idea is a universal concept of which there is only one possible hypotype. If we consider the planning capacity, the machines will be capable of implementing "good projects" from a technical standpoint, without any ability to understand the axiological and ethical context in which their work will be inserted. Consequently, the risks of an ever-increasing spread of "technological

solutionism", without public debate and without political awareness of the decisions to be made, may worsen even more than in the past.

Keywords: background knowledge, Artificial General Intelligence, Molyneux's problem, adversarial attack, latency.

What stochastic parrots tell us about the world of AI

The international success of ChatGPT dates back to December 2022. Following the emotional response elicited by the performances of ChatGPT-3.5, the media outlets of the whole world announced the impending arrival of the AGI, artificial general intelligence, which was going to inaugurate a revolutionary future, entirely incomparable with anything we had experienced in previous history. The trepidation surrounding the creation of the AGI was – and still is – composed of two opposing views. The first perspective imagines a future without any need to work or unsolvable problems, because the machines will be able to take on all operational and intellectual jobs, ranging from labor in industrial and agricultural production, through scientific research, and all the way to the invention of

new technological devices. The second perspective incarnates the downside of the myth of a future without work: mass unemployment, widespread poverty, and cultural and social impoverishment.

On the other hand, the scientific research and technological experimentation aimed towards the development of artificial intelligence has been destined, since the very onset, to the construction of an AGI: Alan Turing's speculations centered on this very subject, and the pioneers gathered at the Dartmouth workshop created the term we now use to label this field of research with the firm belief that human intelligence could be replicated.

In March 2023 I formulated a series of prompts to present to ChatGPT-4 – which at the time had just been released – to verify how close generative transformational AI was to AGI. The request underlying the three prompts that were designed after preliminary research was centered on the creation of an informative essay on the definition, the history and the social implications connected with artificial intelligence. I was therefore asking ChatGPT to give us a family self-

portrait: the «voice» of artificial intelligence was going to tell us the secrets to its identity in the first person and it was going to reconstruct the family tree that it stems from.

ChatGPT-4 began by outlining the essay's table of contents; it then proceeded to compile it, one paragraph at a time, following the summary's articulation. The book was published under the title *Imito, dunque sono?, I imitate, therefore I am?*, which turns Descartes' foundational method of philosophical certainty back into doubt. I believe that the result can teach us some fundamental truths on the state of the art of artificial intelligence and on its distance from what we intuitively classify as intelligence and intelligent behavior. Naturally, the software's mistakes include a cluster of information which is much wider than what emerges from the correctly compiled material. ChatGPT-4 has produced around a hundred «stochastic parrots» in approximately 150 standard text pages. «Hallucinations» and «stochastic parrots» are how artificial intelligence analysts (Bender et al., 2021) have termed content errors which generative

artificial intelligences insert into discourse elaborated in response to users' prompts. At least one stochastic parrot has distanced itself from the type of mistake that a human would make, teaching us to search for what happens in the area of divergence between human error and machine error. In the third paragraph of the seventh chapter, ChatGPT-4 quoted a book which does not exist and the name of an author who also does not exist. The title of the book copies the title of the paragraph which the book is quoted in, and in turn the title of the paragraph appears in the prompt which was imputed to generate the paragraph: *Can artificial agents be autonomous and have free will?* Inventing authors and books is a vice occurring even amongst the best members of the scientific community, and artificial intelligence also sometimes indulges in this habit: the entire last chapter of *Imito, dunque sono?* is an exemplary performance of this practice. But the problem lies in the book's alleged year of publication, which is claimed to be 1995, and yet, according to ChatGPT, Mario Rossi, the purported author of the book, discusses the theses made public by Dennet

in 2003 in *Freedom Evolves* and then supposedly also comments on the 2017 victory of Google's Alpha Go software against the Go world champion, Ke Jie. If Mario Rossi had been capable of such fortune telling, he would have become famous as a prophet, he would have made a fortune by investing in the financial field and he would not have preoccupied himself with obscure metaphysical matters concerning intelligent machines, and we would not be here today discussing any of this. But we do not need this proof by contradiction to notice the discrepancy between the types of stories a student would contrive to hide their unpreparedness on the subject – and the hallucination into which the discourse of the AI falls. Even the world's most ignorant student would avoid attributing the description of ideas and facts which only came to light in 2003 and 2017 to a volume published in 1995, because the configuration of the world that these assertions would have to refer to renders them impossible from the get go. The student frequents this configuration of the world long before needing to consciously thematize it, and the student's

very existence coincides with this primary structure of temporality. Our first patency is that the discourse of ChatGPT does not establish a reference to the world that we inhabit and that we *are* as a precategorial structure to any possible experience.

In 2013 Christian Szegedy, along with a team of researchers from Google and the New York University, conducted an *adversarial attack* on the AI AlexNet, which specializes in image recognition. He selected a group of photographs that the software had correctly classified: a school bus, a temple, an insect, a dog, etcetera. He then inserted *informational noise* composed of colored pixels into the texture of these images. The effect of this interference on the human eye is null: the images that result from this grafting are identical to the starting images. However, for AlexNet this operation is anything but free of consequence: where it used to correctly identify a bus, then an insect, then a temple, then a dog – it now always sees an *ostrich*, and it does so with an even higher degree of confidence as compared to the measure of probability it previously assigned to the correct classifications.

In addition to the parrots, the bird cage of artificial intelligence is thus enriched with another fantastic beast. According to AlexNet, a temple can morph into an ostrich in the blink of an eye, and a school bus can fly away spreading its wings, leaving behind only a few feathers where just an instant before there was a group of children on a school trip. One can easily imagine the kind of risks one would be exposed to if self-driving cars were left to be controlled by artificial intelligence alone. In the world of AI, a bus being overtaken on a highway could fly away as an ostrich or parrot at any moment – and vice versa –, without such an event causing a control response like the one that would ensue in a human blinded by a light or under the effects of a heavy drug. Or should we altogether presume that no world exists at all according to AI?

In the definitions of AGI elaborated by Goertzel and Pennachin in *Artificial General Intelligence* (Goertzel and Pennachin, 2007), further reprised in the following years' editions of AGI conferences, we find *problem solving* as the central notion. Stochastic parrots and ostriches are teaching us to verify that

the adjective general in AGI, if it must have anything to do with natural human intelligence, does not entertain complex relations with *problems*, but entirely unconscious relations with the precategorial stability of the world. Intelligence makes it possible to recognize the relevance of what needs to be thematized against the backdrop of a horizon comprised of all real and possible facts, which remains steadfast and foreseeable. This horizon, which is called «Background knowledge» by the tradition of analytic philosophy and «fungent intentionality» by that of phenomenological philosophy, and which can be assimilated to the «tacit knowledge» of Michael Polanyi, Gilbert Ryle and André Leroi-Gourhan, is the dimension of confidence with the world that artificial intelligence lacks entirely.

Neural networks such as ChatGPT-4 and AlexNet go through a training phase in which they learn how to problem solve, without ever entering into any kind of relation with the world and without being-in-the-world (Merleau-Ponty, 1945). AlexNet learns to associate captions to images, estimating the probability that a certain label within its archive of stored definitions

can be attributed to the imputed pixel patterns. The distribution of colored dots establishes no reference to landscapes and figures that can be observed in the external world: AlexNet actually has no idea that a world exists. In a similar way, ChatGPT measures the probability of what word will have to follow the one that was just printed; for the first sentence of the discourse it uses the dictionary of lemmas that appear in the text imputed by the user. The machine performs a statistical calculation on the chain of the signifier, and completely ignores the signified – or better yet, it ignores the very fact that language should tie some reference to the external world.

For decades, before the paradigm of neural networks became predominant in AI design, theorists and teams of developers imagined to be able to infuse the machine's memory with the archive of Background cognitions available to a human being, or at least to record a sufficient amount of it in order to sustain the focal cognitive operations of the project. Artificial intelligences that «accept» these kinds of suggestions belong to the first constructive paradigm,

shared by Turing and the Dartmouth pioneers: they cannot learn from training; the instructions that guide their behavior are cabled into their code by the programmer. Their entire «mind» is explicitly declared in symbolic language, which therefore also has to contain the memory data that the machine works on in an explicit and unchangeable way. Google's *Knowledge Graph* project, launched in May 2012, might be one of the most famous examples of this heroic project, aiming at an infinite task: eight years later, in May 2020, the system harbored 500 billion facts on 5 billion entities. Despite the huge amount of data, correlated in cognitive structures, the task to form a base of minimum information to allow access to a world of meanings and practices, to an inhabitable universe such as the human one, had not even begun. The consolation here is that, on the other hand, the service made available in the Google answers listing page offers an interesting support to the serendipitous search method.

Another typical error of the pioneers has been the adhesion to a naïve empiricism, which keeps fueling

AI projects even in the neural networks era. Marvin Minsky and Seymour Papert, back in the spring of 1966, professed their belief to be able to build a complete visual system during the summer semester. The principle on which AlexNet operates, upon which recognizing image content coincides with the ability to recognize objects' outlines, does not greatly differ from the conviction that a camera and a few algorithms with explicit image reading rules suffice to mimic the perceptive system of vision. If the pioneers and their successors had been awareof the very philosophical matter of *Molyneux's problem*, which dominated the cultural debate between the 17th and the 18th century, they would have approached the matter more cautiously.

William Molyneux, astronomer and physicist of the Trinity College in Dublin, posed this question to John Locke in a letter in 1688: let us imagine a man who was born blind, who has learned to distinguish objects by touch, and can therefore identify a cube and a globe based on the sharpness of the former and the convexity of the latter. If one day this individual

were suddenly given the ability to see, would he be able to distinguish the two geometric shapes just by observing them, without using touch? Locke contributed to bringing this problem to the attention of the intellectuals of the time by including it in the second edition of his *Essay Concerning Human Understanding*, thus making it the most important question in 18th century knowledge theory, according to Ernst Cassirer. One interesting aspect of the debate between the supporters of the affirmative answer and the defenders of the negative thesis is that in 1728 the matter could be verified experimentally, due to the first cataract surgery performed by William Cheselden on a fourteen-year-old boy who had been blind since birth (Cheselden, 1728). The boy had not only been unable to identify the cube and the globe, but he also had the feeling that everything touched his eyes or was placed upon their surface and could not distinguish shapes and evaluate distance nor depth. In other words, the boy only saw stains of color which mixed up within each other, as if they were internal to his sensory organ. We could simplify this by saying that he only

perceived informational noise, with no access to the objects: neither cube, nor globe, neither school bus, nor ostrich. One cannot recognize anything through the mere mechanical operation of exposing images to the eye, followed by the abstraction of the elements recurring in similar configurations. Categorization is an essential operation to move from stains of color in the perceptive apparatus to the natural ease with which we position a world of objects and events outside of ourselves.

General intelligence relates to being able to drive a school bus along a road by a temple without expecting the temple to take off and fly away with a flock of ostriches, and without expecting the school bus to fly over the temple in a matter of seconds – it pertains to the articulation between the Background and the events which are possible within it, rather than the ability to solve problems, write literary *pastiches* for the Goncourt Prize or create hyper realistic images resembling photographs, mixing artistic styles which have disappeared decades or centuries ago. The fears and hopes connected to AGI should therefore be

postponed to future times, which we cannot currently foresee (unless, maybe, you are the Mario Rossi described by ChatGPT-4). We have no idea how to engineer the primary structure that opens the world and the self-conscious subjectivity which inhabits it: therefore, on our horizon there are no lingering digital subjects capable of elaborating autonomous projects surrounding the destiny of the universe, the Earth and humanity, nor their own survival on the servers of the company that created them. At least as long as the current paradigms of AI modelling remain in place, we will not have machines with autarkic world views, nor with revolutionary aesthetic and moral taste. But this does not mean underestimating the contribution that neural networks have given to the lifestyle of western humanity in the past decades. Every day we use artificial intelligence devices, symbolic or neural, which have revolutionized everybody's work- and daily life, such as Google, spam filters, social medial profiling systems, facial- or fingerprint identification, etcetera. Generative transformational AIs have opened many thought provoking perspectives on the fields of

production that were traditionally solely assigned to the specificity of the human spirit. Amongst these, I would like to outline an exemplifying analysis, centering on the originality of individual creativity.

In §49 of the *Critique of Judgement*, Kant qualifies the aesthetic idea as a universal concept, which has *only one* possible hypotyposis, the work of art which incarnates it, but which feeds an endless process of interpretation – since no language and no discourse will ever be able to enclose its overflowing capacity to signify into one exegetical formula. The performances of ChatGPT, but maybe in a more effective manner those of figurative generation software such as Midjourney or Dall-e, put in question the solitude of the Kantian hypotyposis, because they respond to users' requests with the production of variants of past works of art, or by re-elaborating the style of a certain author or epoch, or by combining the tastes and mannerisms of different artists or periods. As Arielli and Manovich point out, the result of these operations seems to bring to light a «latency» which thus far remained collapsed in the choices made by

the author (or publisher, or client) in the final version of the work of art (Arielli and Manovich 2021-24, 6). Artificial intelligence traces a set of patterns within the phenotypes of human creativity that escape the awareness of artistic movements and of the forms that are passed down and re-elaborated by academies. It can therefore retrace a set of perceptive rules that belong to the almost ineffable dimension which allows us to intuitively or emotionally recognize the imprint of a certain author or period, which usually does not surface to explicit awareness. But this opportunity is also accompanied by a question surrounding the ontology of these variations and the worth we can attribute to them. If this question has an obvious relevance for what Nelson Goodman (Goodman, 1968) terms *autographic* arts, like painting and sculpture, it is no less important for the *allographic* arts. What is the completion of the last two movements of Schubert's Eighth Symphony, specifically famous for being *Unfinished*? What meaning could it aim to express for us, for the history of music, for the understanding of Schubert? What is the ontological statute of a

fragment of music sheet that should have come to life two centuries ago? What is this reparation, today, of a miscarried act? What right do we have to impose on the unfinished nature of Schubert's symphony the character of a defect, a deficit? And most of all, what right do we have to put a remedy to it today? The return of the removed always poses problems of meaning and evaluation, requires legitimation and validity criteria, and needs a permit to be integrated into the habits, the balance and the stability of current practice. AI lifts out of the unknown an area of the unconscious which is intrinsic to art, hence a sphere of extraneity in what most belongs to mankind, in the technical essence of its homination process, bringing with it anxiety, fascination, and trepidation for what belongs to the spontaneous beauty of existence – but also the stupidity, bitterness, Kitsch of what has been rejected, and which in its abandonment suffers the violence of the state of exception, the liminality of what is accepted only because it is outcast, because it has been and it is *other*, artificial, spurious, anonymous.

These are all questions that we should strive to

find an answer to, so that, at least we humans, can continue to have a world.

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7

AI AND THE POLITICS OF POWER: IMPLICATIONS FOR SOCIETY AND THE ARCHITECTURE OF SPACE

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AI AND THE POLITICS OF POWER: IMPLICATIONS FOR SOCIETY AND THE ARCHITECTURE OF SPACE

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Abstract

AI and digitalisation are fast transforming domain areas, including that of architecture. These technologies enhance efficiency, productivity, and decision-making while improving urban living through better architecture design, management and service delivery. However, the rapid adoption of AI and digital technologies also introduces profound social and ethical challenges. This chapter explores these

challenges and argues for a more Responsible AI by considering Responsible Innovation in architecture in order to proactively address the ethical concerns that arise.

Keywords: AI, Power, Politics, Space, Society

7.1. Introduction

AI and the digitalisation process present a significant impact on society in a number of ways including socially, politically and culturally. For the most part, this impact has been largely positive, with AI and the digitalisation process enhancing various sectors such as industry, education, healthcare, entertainment and many other domain areas. For instance, in the industry sector, AI technologies have boosted efficiency and productivity, with AI-driven predictive analytics enabling businesses to optimise their operations (Peres, 2020). Additionally,

AI's capacity to process vast data volumes facilitates improved decision-making and efficiencies.

The education landscape is also changing due to AI and digital technologies. As Tahiru (2021) suggests, the introduction of AI is transforming education in various ways which include automating administrative tasks and providing smart content and intelligent tutoring systems. Similarly, AI's application in healthcare has been beneficial, leading to the leveraging of large data-sets for more accurate disease diagnosis and timely treatments (Han et al, 2020). Overall, the widespread adoption of AI and digital technologies is driving significant advancements across various domains, marking a pivotal shift in how societies function and thrive in an increasingly digital world.

7.2. AI Architecture and Space

The impact of AI and digitalisation also extends to architecture and the design of society's space. This is done through better efficiencies and optimisation in for example the development of smart cities. As noted

by Allam & Dhunny (2019), the integration of sensors and big data, propelled by AI advancements, is unlocking new possibilities for architectural innovation and environmental management. These technologies not only boost creativity and sustainability but also enhance the functionality of urban spaces, enabling more effective utilisation of physical space. This transformation is particularly evident in the development of smart cities, where AI-driven technologies facilitate smarter urban planning and more efficient resource management. According to Ortega-Fernández (2020), AI's application in urban settings has led to improved infrastructure management, transportation, and public services, contributing to more sustainable and efficient living spaces.

Beyond these improvements, the use of AI in architecture also addresses broader societal challenges, such as sustainability, accessibility, and adapting to changing demographic needs. AI-driven tools can analyse environmental data and optimise building designs to minimise carbon footprints (Wu et al, 2022), making sustainable practices more

accessible and cost-effective. Veisi et al. (2024) have demonstrated how AI could be leveraged in urban planning and climate adaptation, noting a high level of precision in environmental modelling enabled by AI technologies. Furthermore, AI's predictive capabilities are essential for anticipating demographic shifts (Louadi, 2024) and urban growth, allowing for the design of scalable and flexible buildings that can adapt over time.

The integration of AI in architecture also means that architects can now utilise vast amounts of data for predictive analytics in smart city design, as highlighted by Cong & Inazumi (2024). However, the rise of AI-driven architecture brings with it social and ethical concerns, with Liang et al (2024) identifying issues such as job loss, data privacy, data security, data transparency, decision-making conflict, acceptance and trust, reliability and safety, fear of surveillance, and liability. This shift challenges conventional norms of architecture and prompts a re-evaluation of architecture through the prism of responsible AI and a consideration of ethics in the digital age. The evolving

role of architecture in this context underscores the need to balance technological advancements with the safeguarding of societal space with respect to personal and environmental concerns.

7.3. Social and Ethical Concerns

While AI's integration into architecture offers numerous benefits and transformative impacts on societal space and its architectural design, the application of AI-driven architecture introduces complex ethical and socio-cultural concerns which potentially include power dynamics, politics, people, and space. Such ethical concerns can give rise to tensions among various social, economic, and political entities, ultimately affecting people, spaces, and the environment.

7.4. Space and the Politics of Power

Loss of Control of Personal Data and Privacy

The deployment of AI and digital technologies in smart cities for example brings to the forefront significant ethical challenges, particularly concerning privacy and data security. The extensive use of technologies such as cameras and mobile devices in urban spaces, highlighted by Rizi & Seno (2022), creates vast datasets that can infringe on individual privacy. Mobile devices are commonly used for transactions and accessing various city services, such as paying council bills, scheduling hospital appointments, and using public transportation (Al Nuaimi et al, 2015), raising critical questions about data protection, access, and management.

Furthermore, the integration of AI technologies in smart cities increases the surveillance and tracking capabilities, significantly impacting citizens as they navigate their daily lives using smart devices

(Eigenraam & Rothkrantz, 2016). This pervasive monitoring often occurs without explicit consent from individuals, resulting in a considerable digital footprint and limited control over personal data. The typical lack of options to opt out of data collection by digital services exacerbates concerns about data sharing with third parties.

Third Parties and Tech-giants

Private companies specialising in AI and digital technologies often play a crucial role in developing smart cities and enhancing their operational efficiency, therefore influence the architecture of space. These organisations frequently require access to citizen data, while city authorities may share this information without fully informing citizens about how their data is managed or utilised (Mercille, 2021). This scenario underscores significant concerns regarding the politics of power and data governance.

Tech-giants hold near-monopolistic influence in the digital economy, which grants them considerable power over civic engagement. This control is evident in their

ownership of communication platforms that facilitate citizen interaction and collective action (Moore, 2016). Kwet (2019) points out the monopolistic power of entities like US-based tech-giants, which extends to the architecture level of the digital ecosystem through their provision of software, hardware, and network services. He argues that such power allows the tech-giants to dominate economically, exert imperial control, and propagate global surveillance capitalism. Such power has an impact on how architecture is and can be implicated in aspects of power and space. For instance, the influence of these tech-giants extends into the very fabric of urban planning and infrastructure development, where smart city projects often depend heavily on the technologies and platforms they provide.

This dependency on a few powerful players for critical infrastructure and services can lead to a form of digital dependency where cities and communities may have limited control over their own data and the tools they use to manage it. The design of public spaces increasingly incorporates AI technologies provided by these corporations, potentially embedding corporate

interests into the layout and function of these spaces. This raises significant concerns about privacy, data sovereignty, and the equitable access to public spaces that are increasingly mediated by private technology.

Furthermore, data harvested in smart environments can be strategically used to shape behaviours and regulate access in ways that prioritise profit over the public good. Zuboff (2019) has argued that data derived from human experiences, treated as a freely available resource, is utilised to predict behavioural trends. This predictive analysis is then used to customise products and services to meet both current and prospective needs, a practice that significantly benefits entities engaged in surveillance capitalism.

Such practices underscore a profound power dynamic with extensive architectural implications, impacting everything from the management of pedestrian traffic to the accessibility of city services. Gillespie (2002) demonstrated that aspect of power dynamics in architecture which can result in social divisions and inequalities. This influence often occurs without sufficient oversight or meaningful input from

the communities affected. The integration of these data-driven technologies into public spaces not only shapes urban life but also raises critical questions (Chang, 2021) about who benefits from these configurations and at what cost to individual privacy and community autonomy. This brings in the aspect of how the personal becomes the political in architectural design.

The Personal and the Political

Inclusion and engagement are an important aspect as AI technologies can alter conventional notions of space for individuals inhabiting the spaces. In his preface, Frey (2010) has opined that:

what all agents of change need to do is to learn how to move design out of its economic function and into a political frame. But for this to be possible the current nature of that frame needs to be understood. For people outside design practice, the potential of design transformed requires exposure, as does the current designing nature of the political.

This also underscores the concept that ‘the

personal is the political', which can be extended to emphasising how individual interactions with AI-driven environments reflect and influence societal structures and debates. The "the personal is the political" can be implicated in discussions about the impact of AI on individual lives within their broader societal and political contexts. The phrase underscores the idea that personal experiences, particularly those shaped by AI technologies, are not isolated incidents but reflections of larger political implications. When AI influences aspects of daily life, such as privacy, control accessibility, and autonomy, security among others, the technologies subsequently become implicated in larger political and ethical issues. For example, the use of AI in surveillance applications as discussed by Heimann & Hübener (2023) can raise significant concerns about individual privacy rights, which are inherently political. Similarly, how AI enhances or impedes accessibility in architecture design can reflect societal values or biases, linking personal experiences to wider political discussions. In this context, individual interactions with AI systems becomes interconnected with broader

political debates, as each personal experience of AI and other digital technologies becomes a reflection of and a contributor to the governance, policies, and cultural norms of the digital era.

As such, addressing the above challenges requires a nuanced approach to architecture design and space that incorporates transparency, citizen participation, and robust legal frameworks to protect individual rights and ensure equitable outcomes. One way that this may be achieved is through a consideration of the concept of Responsible AI in architecture which can contribute to fostering a more inclusive and engaging process in the deployment of AI technologies and design of space that can safeguard against practices that undermine public trust and social justice.

7.5. Embedding Responsible Innovation in Architecture and AI

Responsible Innovation (Responsible Innovation) can play a crucial role in addressing ethical concerns of AI and architecture, ensuring that the AI technologies

used in architecture are both socially beneficial and ethically sound. By adhering to the principles of RI, stakeholders in AI and architecture are compelled to consider not just what can be done technically, but what should be done ethically and morally. This involves a thorough assessment of right and wrong in every phase of a technology's lifecycle, from design and development to implementation and maintenance. For architecture, this means creating spaces that are not only innovative and efficient but also accessible and respectful of the cultural and social contexts in which they exist.

In practice, RI requires that AI systems used in architecture be inclusive, catering to a wide array of needs and actively avoiding the reinforcement of existing biases or inequalities. This goal can be achieved through the implementation of the AREA framework, which includes Anticipation, Reflection, Engagement, and Responsiveness, as outlined by Owen et al (2013). For instance, in the Anticipation phase, employing AI in architectural design will require a thorough analysis of both the intended and potential

unintended effects of the AI technologies. This analysis should cover possible outcomes, risks, and uncertainties, a method detailed by Owen & Pansera (2019).

Reflection will manifest in how architects and designers will critically assess their processes of design, examining the underlying assumptions, values, and implications of their designs. The Engagement aspect seeks to expand the decision-making process, involving not just architects but also the broader public. This inclusive approach facilitates deeper, more democratic deliberations about the design and utilisation of spaces, ensuring that diverse perspectives are considered.

Lastly, the Responsiveness dimension of RI guarantees that the insights derived from anticipatory and reflective efforts are effectively applied to refine the integration of AI in architectural practices. This involves adapting AI's core objectives, processes, and ethical frameworks to ensure they align with societal values and regulatory standards. Through these integrated approaches, AI in architecture can

be responsibly harnessed to serve and enhance architectural spaces while addressing critical ethical concerns.

The integration of RI into the fields of architecture and AI is essential for addressing the multifaceted challenges posed by the politics of power in AI deployment. As AI technologies continue to reshape architectural spaces, the need for a responsible AI that prioritises ethical considerations, inclusivity, and stakeholder engagement becomes increasingly evident. RI provides this framework, advocating for AI technologies that are not only innovative but also socially responsible and aligned with the public good. By applying RRI principles, architects can ensure that their designs do not merely serve the interests of a powerful elite but are attuned to the broader societal impacts, promoting fairness and equality.

7.6. Conclusion

In conclusion, there is no doubt that AI and the digitalisation process have brought about

transformative change across different domain areas including that of health, education, business, and architecture among others. The transformation is evident in various ways and includes enhancing efficiency, productivity, and decision-making. AI and related other digital technologies are also significantly contributing to improved urban living through better infrastructure management and service delivery. However, this technological evolution is not without its challenges. The widespread adoption of AI and digital technologies continues to raise critical questions around ethical concerns in addition to their implications for privacy, data security, and the socio-cultural fabric of society. These technologies have inadvertently led to increased surveillance and tracking, infringing on individual privacy and creating a digital divide that excludes those without access to AI and other related technologies. Furthermore, the global discourse on AI is predominantly shaped by the perspectives of the Global North, often neglecting the unique needs and ethical considerations of the Global South, leading to a form of epistemic injustice.

The emergence of surveillance capitalism, data colonialism, and the considerable power presented by technology giants underscores the urgent need for the ethical governance and responsible implementation of AI and digital technologies. Responsibility in AI and the digitalisation process has the ability to ensure the promotion of equal access to the technologies and also to ensure sustainable growth while at the same time anticipating and finding solutions to ethical concerns of the technologies as they become mainstream.

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Ingrid Campo-Ruiz's research focuses on architecture and artificial intelligence, and their social and environmental impact. She develops strategies to enhance social inclusion and preserve the environment through machine learning.

Campo-Ruiz is a practicing architect, researcher, environmental advocate, and artist. A European Commission Marie Sklodowska-Curie Actions Individual Fellow at KTH School of Architecture, Stockholm, she is also the founder of Campo Ruiz Architect, LessWasteArchitecture, and #whylesswaste.

Campo-Ruiz earned an International PhD in Advanced Architectural Projects from the Madrid Polytechnic University UPM, ETSAM, Spain, received a Master of Science in Advanced Architectural Design from Columbia University, GSAPP, New York, where she also did research at the Advanced Architectural

Research concentration. She studied architecture at ETSAM and the Copenhagen School of Architecture, Denmark.

Campo-Ruiz has been awarded first prize at the European Commission Marie Sklodowska-Curie Actions Grant International Conference, the William Kinne Fellowship from Columbia University, and two Caja Madrid Foundation Fellowships. She is a TEDx speaker and has exhibited her work in Madrid, Brussels, Stockholm, Copenhagen, and Turin.

Katja Grillner is professor in critical studies in architecture and the head of department at the Department of Architecture, KTH (2019-). Among previous leadership assignments at KTH she has served as Dean of faculty (2015-2019), deputy head of department (2013-2015) and deputy head of the ABE-school (2009-2011).

Her research into architecture and landscape perform aesthetic, historical and literary investigations from the perspectives of poetic practices and critical theory. In her teaching she has together with her colleagues contributed to a presence of courses or course components concerning intersections between architecture, feminism and gender in the architecture program and in extra curricular courses at the school.

Grillner has initiated and participated in the formation of several groups and contexts that has

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Peter Lynch, architect and educator, has taught at Harvard Design School, Columbia University, Rhode Island School of Design, City College of New York, Parsons School of Design, Dalhousie University, Penn State University, and KTH, where he was guest professor from 2015-2020. He headed the graduate architecture department at Cranbrook Academy of Art from 1996-2005. His upcoming book *Fragments and Coherence* will be released by Skira Editore in 2025.

Dr. Mats Nordahl is a complex systems and data science researcher. He is the founder of three startup companies working on applications of AI, in particular relating to the interpretation of sensor and visual data. He has also been involved in a range of interdisciplinary collaborations with artists and architects/designers, resulting in exhibited work and new developments in higher education. Lynch and Nordahl's current research project, "Timescape Garden," an urban biodiversity preserve in Norrköping, is funded by Vinnova, Sweden's innovation agency.

Valarie Waswa is a Legal Fellow at KICTANet, Kenya's leading ICT Policy think tank, where she focuses on digital rights advocacy, policy analysis, and supporting the development of regulatory frameworks that promote internet governance and data protection.

Her expertise spans AI ethics, digital rights, and the ethical governance of technology. With a background in law and policy, Valarie has previously worked as a Program Coordinator at She Goes Digital KE, supporting programs that seek to bridge the gender digital divide in rural communities in Kenya. Before that, she served as a Digital Youth Envoy at the International Telecommunications Union (ITU), The UN Agency for ICTS. Her role at ITU included promoting meaningful youth engagement in global digital development and policymaking.

Valarie holds a Bachelor of Laws degree from the Catholic University of Eastern Africa and is an Advocate in training. Having shared her insights at national, regional, and international forums, Valarie has become a recognized voice in critical discussions around internet governance, digital development, and technology for good (Tech4Good).

Valarie's professional journey is guided by a deep commitment to tech justice and the belief that technology when used responsibly, can be a powerful force for good.

Paolo Bottazzini has a background as a philosopher and is co-founder of the companies Pquod, SocialGraph and VentunoLab, specializing in web communication, SEO, social media and Big Data. Professor of social network analysis at the Department of Philosophy of the University of Milan, he has taught on the topics of digital communication for the Polytechnics of Milan and Turin. Among the published volumes: Anatomia del Giudizio Universale. Presi nella Rete (Mimesis, 2015) and Idee Virali (il Mulino, 2018, with Emanuele Arielli). As a journalist he has collaborated as a scientific communicator with Linkiesta, pagina99, Gli Stati Generali, Nova de Il Sole 24 Ore.

Dr. Kutoma Wakunuma is a researcher and academic with interest in the field of AI and digital technologies, ethics and responsible innovation. Her research interests and expertise primarily focus on the ethical implications of current and emerging technologies on modern society including both the Global North and the Global South. She has contributed significantly to understanding how technology impacts society, analysing the consequences of technology deployment and exploring responsible and ethical approaches to technology development and policy. Dr. Wakunuma is a regular speaker on issues related to understanding how technology impacts society. She has been invited to convene and speak at high level international forums, including the United Nations Academic Impact Commission on the Status of Women Event discussing the theme of Strengthening Gender Equality in Generative AI; the UNs Science

Summit of the General Assembly and the UNs Internet Governance Forum where her insights contribute to global discussions on technology and society. She also acts as an Ethics Advisor to the European Commission.

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